St. George's University
Grenada, West Indies

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary & Renal

Lecture: CPR 29

Inflammation (Flipped Class Session)

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Objectives

State the importance of inflammation	SOM.MK.I.BPM1.3.CPR.1.HCB.1287
List the two major categories of inflammation.	SOM.MK.I.BPM1.3.CPR.1.HCB.1288
Describe the three major components of the acute inflammatory exudate.	SOM.MK.I.BPM1.3.CPR.1.HCB.1289
Discuss the mode of formation of an acute inflammatory exudate.	SOM.MK.I.BPM1.3.CPR.1.HCB.1290
List the vasoactive mediators involved in various stages of the acute inflammatory response.	SOM.MK.I.BPM1.3.CPR.1.HCB.1291
List the chemical mediators involved in various stages of acute inflammatory response.	SOM.MK.I.BPM1.3.CPR.1.HCB.1292
Describe the three morphological types of acute inflammation.	SOM.MK.I.BPM1.3.CPR.1.HCB.1293
Give an example of each of the morphological types of acute inflammation.	SOM.MK.I.BPM1.3.CPR.1.HCB.1294
Define the terms “pus”, “abscess” and “pyogenic bacteria”.	SOM.MK.I.BPM1.3.CPR.1.HCB.1295
Describe the three major components of the acute inflammatory transudate.	SOM.MK.I.BPM1.3.CPR.1.HCB.1296
Discuss the mode of formation of the acute inflammatory transudate.	SOM.MK.I.BPM1.3.CPR.1.HCB.1297
List and describe the cardinal signs of inflammation.	SOM.MK.I.BPM1.3.CPR.1.HCB.1298
List and describe the three possible outcomes of acute inflammation.	SOM.MK.I.BPM1.3.CPR.1.HCB.1299
Describe the cell types that are predominant in acute inflammatory infiltrates.	SOM.MK.I.BPM1.3.CPR.1.HCB.1300
Describe the main factors that determine the outcome of an acute inflammatory condition.	SOM.MK.I.BPM1.3.CPR.1.HCB.1301
Describe the role of granulation tissue in wound healing and resolution of acute inflammation.	SOM.MK.I.BPM1.3.CPR.1.HCB.1302
Describe the cell types that are predominant in chronic inflammatory infiltrates.	SOM.MK.I.BPM1.3.CPR.1.HCB.1303
Describe the typical outcome of chronic inflammation.	SOM.MK.I.BPM1.3.CPR.1.HCB.1304
Define granuloma.	SOM.MK.I.BPM1.3.CPR.1.HCB.1305
Describe 2 different types of granuloma formation.	SOM.MK.I.BPM1.3.CPR.1.HCB.1306
Identify the predominant cell type in granulomas.	SOM.MK.I.BPM1.3.CPR.1.HCB.1307
Compare and contrast granulation tissue vs. granuloma.	SOM.MK.I.BPM1.3.CPR.1.HCB.1308

Recommended Reading

For this lecture:

*Incorporate lecture material on Blood

In particular - the white cell lineage

*Lecture Handout

*Images:

Wheater's Basic Pathology – 5th edition

Robbins Pathological Basis of Disease – 8th edition

Robbins Basic Pathology, Chapter 3, 57-96 on clinical key:
https://www.clinicalkey.com/student/content/book/3-s2.0-B978032335317500003X#hl0000855

Outline

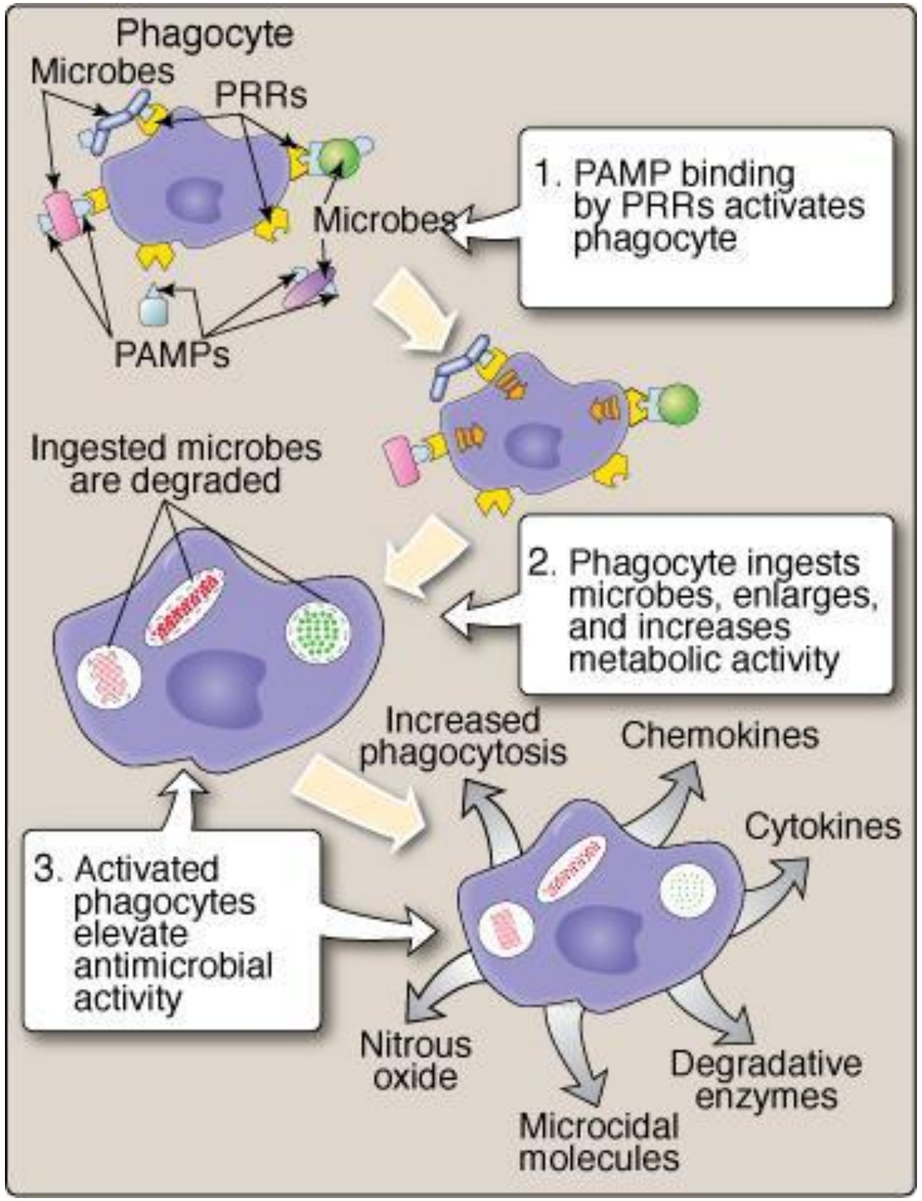
- What is inflammation
  - Difference between acute and chronic
- Acute inflammation
  - Sequence of events
  - Cells associated and when they are predominant
  - Cardinal signs
  - Sequence of events details and mediators involved
  - Specific types
  - Possible outcomes
- Chronic inflammation
  - General description
  - Cells involved
  - Different types

Case presentation – Case 1

A 46-year-old man, Mr. Smith, comes to the emergency department because of a 2-hour history of fever, nausea and vomiting, and a 3-day history of poor appetite, lethargy, and a painful swelling between his thighs. He has a history of type-2 diabetes mellitus to which he is non-adherent to diet or medications. His blood pressure is 140/89 mmHg, pulse is 102/min, respirations are 23/min, temperature is 104°F (40 ° C) and random blood glucose is 550 mg/dL (normal range: 70 – 110 mg/dL). General physical examination shows an anorexic, febrile man in severe painful distress.

Innate Immunity

- Components are always in activated or near-activated state
- Fast response
- Responsible for inflammation
- Usually enough to solve the issue
- Genetically programmed to recognize broad classes of pathogens
    - Receptors are ***“hard-wired”*** genetically - **pattern recognition receptors (PRRs)**
    - Pattern recognition receptors e.g. toll-like receptors
    - Detect **pathogen associated molecular patterns (PAMPS)**
- Cells involved are mostly myeloid lineage
    - Granulocytes e.g. neutrophils
    - Monocytes and macrophages
    - Dendritic cells e.g. Langerhans cells in skin
    - Natural killer cells (lymphocytes)



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Types of inflammation

Acute

- Rapid onset
  - Typically, within minutes
- Short duration
  - Hours or a few days
- Characteristics
  - Edema (swelling)
  - Emigration of leukocytes (Neutrophils)
  - Redness
  - Pain

Chronic

- May follow acute inflammation if the stimulus persists
- May also be an insidious, low-grade response without acute reaction
- Long duration
  - Weeks or months
- Characteristics
  - Macrophages & Lymphocytes
  - Proliferation of blood vessels
  - Fibrosis
  - Tissue destruction

When the process is excessive or defective, it can cause disability e.g. rheumatoid arthritis, atherosclerosis, and lung fibrosis, as well as life-threatening hypersensitivity reactions to insect bites, drugs, and toxins

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Acute - cardinal signs

External manifestations of Inflammation (Celsus):

1. **Redness** (rubor) - hyperemia
2. **Swelling** (tumor) - fluid exudation and hyperemia
3. **Heat** (calor) - hyperemia
4. **Pain** (dolor) - release of bradykinin and PGE_2
5. **Loss of function** (functio laesa) – combined effects, mainly swelling and pain (Virchow)

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Role of Mediators in Acute Inflammation

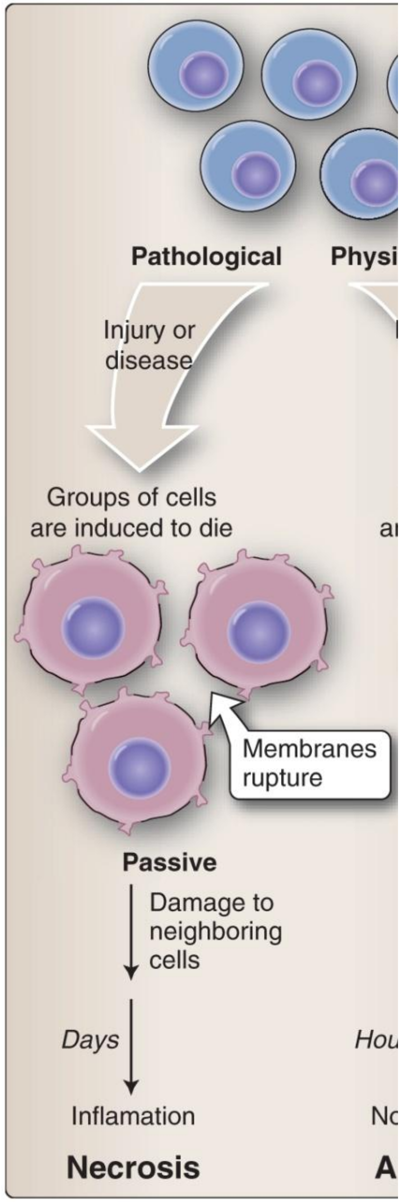
Role	Mediators	Associated cardinal sign
Vasodilation	Prostaglandins, nitric oxide, histamine	Redness, heat
Increased vascular permeability	Histamine, serotonin, bradykinin, leukotrienes	Redness and Swelling
Chemotaxis, leukocyte recruitment & activation	Cytokines (TNF, IL-1), chemokines, C3a, C5a, leukotrienes, bacterial products & peptides	Swelling
Fever	Cytokines (IL-1, TNF), prostaglandins	Not a cardinal sign
Pain	Prostaglandins, bradykinin	Pain
Tissue damage	Leukocyte lysosomal enzymes, reactive oxygen species, nitric oxide	Loss of function

Cell Death

Necrosis

- Pathological
- Acute cell injury
- Cell unable to maintain homeostasis
- Cell swelling
- Loss of plasma membrane integrity
- Cell contents released
- Surrounding tissue damage
- Inflammation

Necrosis- Principle outcome in many injuries
Eg. Ischemia, toxins, infections & trauma



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Feature	Necrosis	Apoptosis
Cell Swelling	+++	-
Cell Shrinkage	-	+++
Damage to plasma membrane	+++	-
Plasma membrane blebbing	-	+++
Aggregation of chromatin	-	+++
Fragmentation of nucleus	-	+++
Oligonucleosomal DNA fragmentation	-	+++
Random DNA degradation	+	-
Caspase cascade activation	-	+++

Morphological Differences

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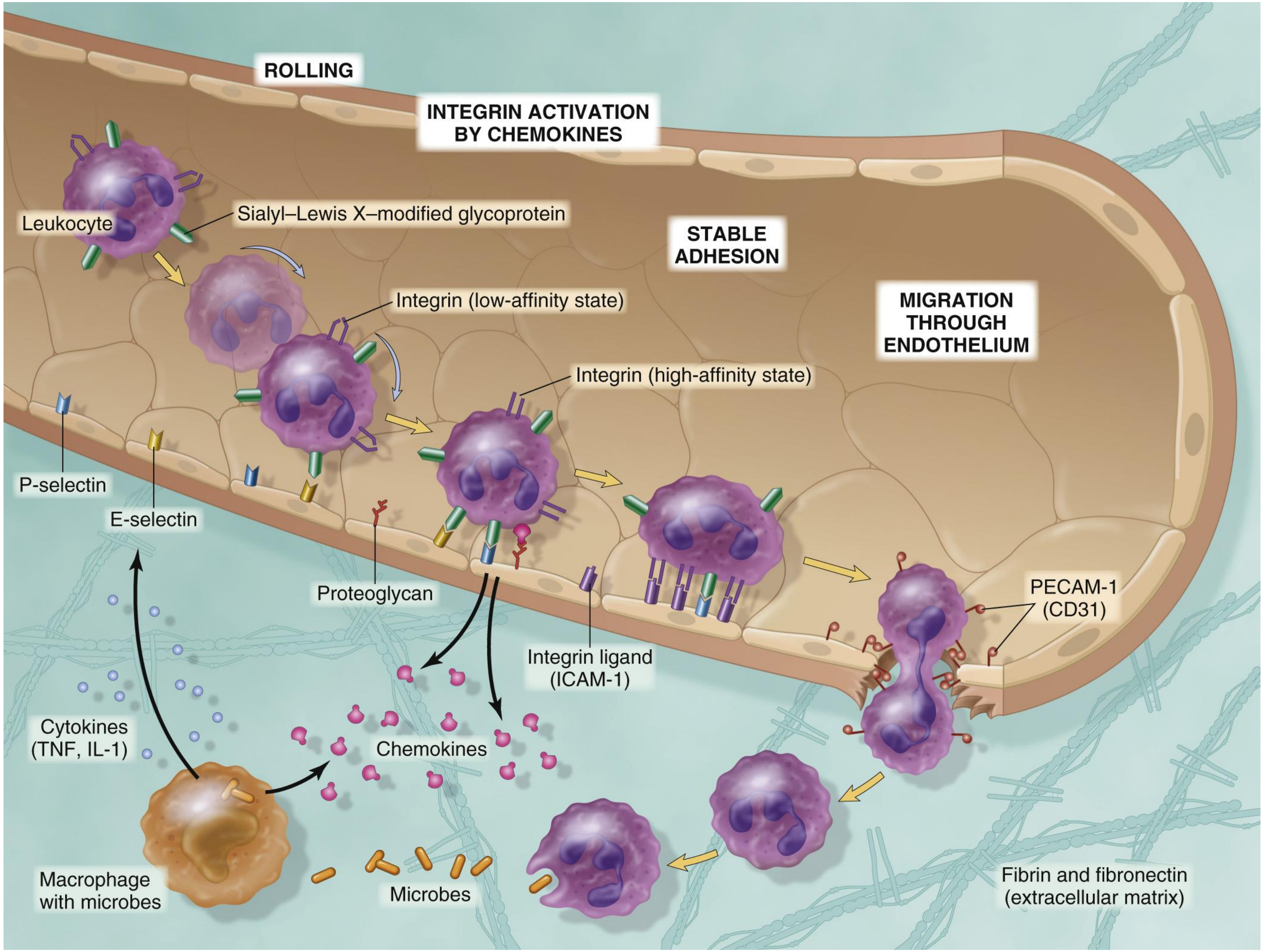
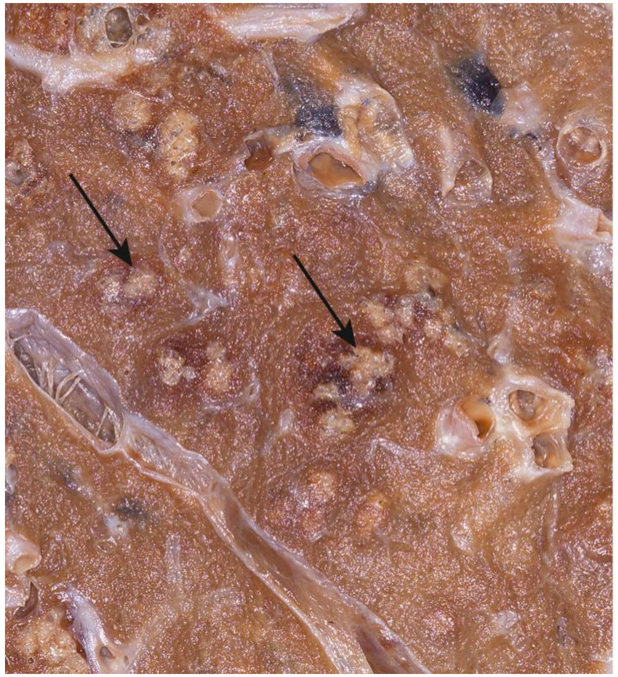
Acute inflammation - Leukocyte recruitment and migration

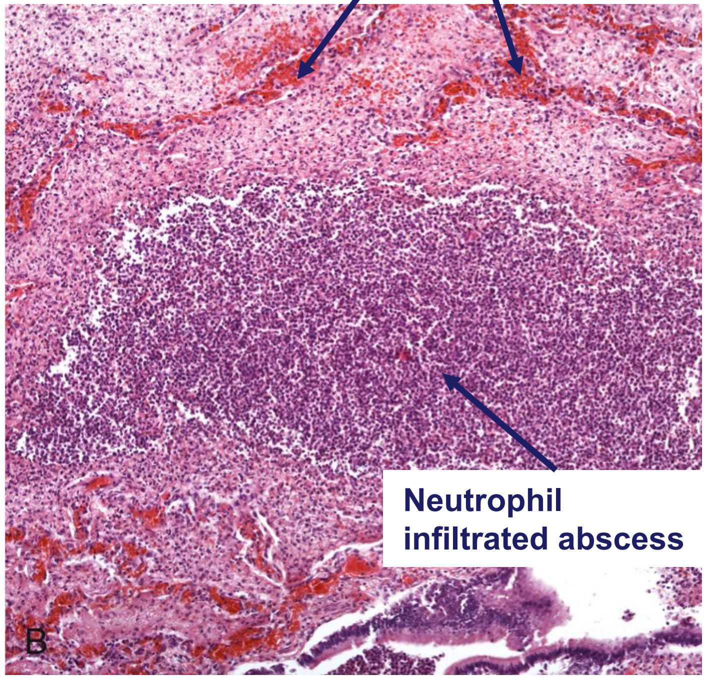
- *Margination*
- *Rolling*
- *Adhesion to endothelium*
- *Migration across the endothelium*
- *Migration in the tissues*



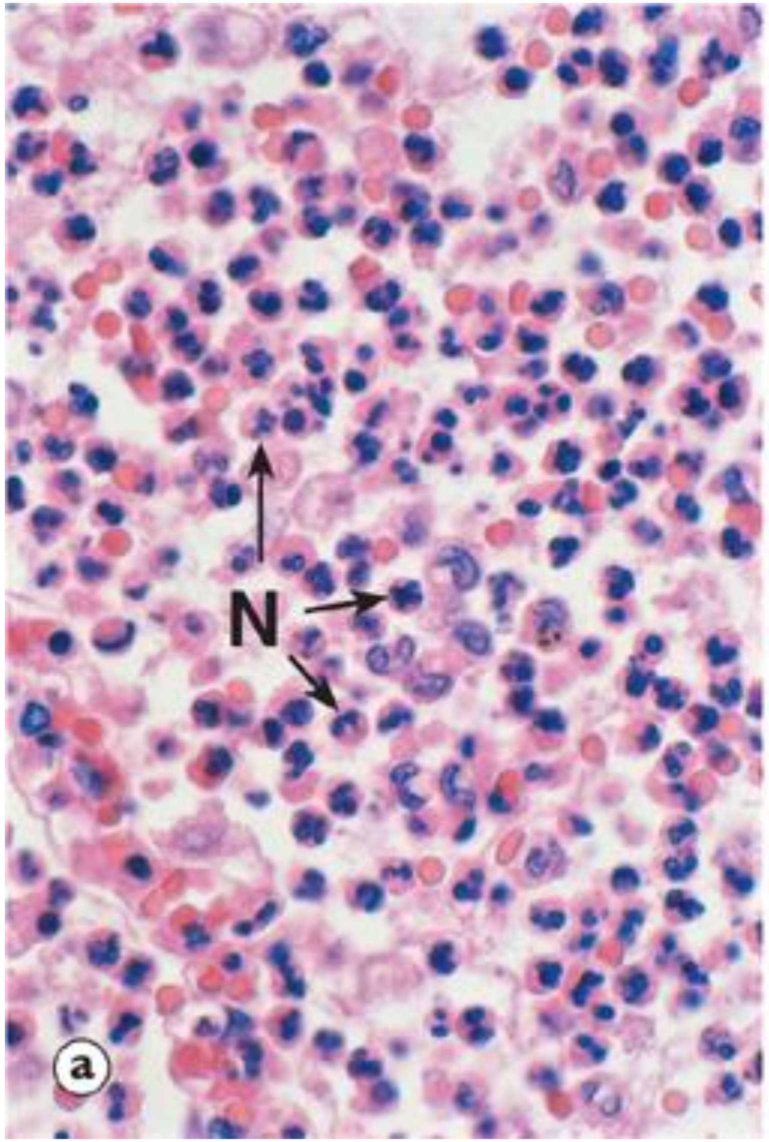
Morphological Types of Acute Inflammation - Suppurative or purulent (pus-containing, exudative) inflammation

- Bacteria that promote purulent inflammation are called **"pyogenic"** bacteria
  - Staphylococci
  - Streptococci such as S. pyogenes and S. pneumonia
- Exudate in purulent inflammation is rich in **neutrophils, liquefied debris**, and edema – Pus
- An ***abscess*** (arrows) is a circumscribed collection of pus
- Common examples:
  - Lobar pneumonia
  - Bronchopneumonia (image)
  - Acute appendicitis

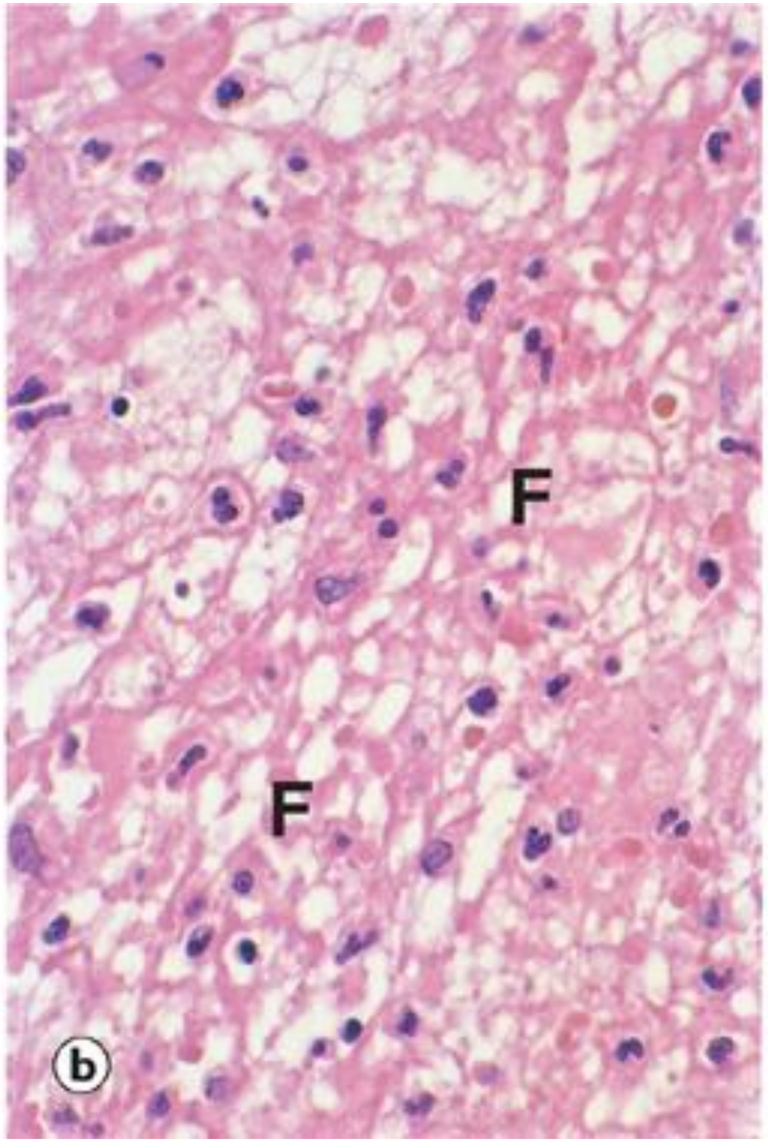




Suppurative vs. Fibrinous Acute Inflammation



Purulent - Exudate with neutrophils (N) as main component



Fibrinous - Exudate with fibrin (F) as main component and few neutrophils

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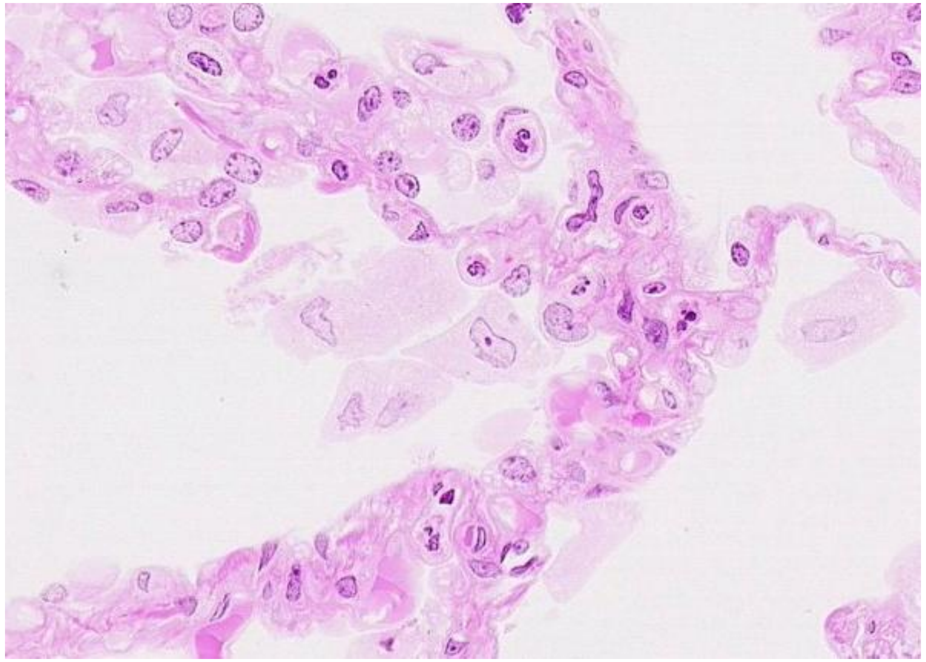
Macrophages

- Specialized cells that migrate to the site of infection and ingests microbes or cellular debris **(Phagocytosis)**
- Give rise to **antigen presenting cells** and multinuclear giant cells



Case presentation – Mr. Smith (46yoa) … some time later …

Mr. Smith returns 1- month later with additional symptoms.

He has been experiencing fever, coughing, night sweats and chest pain for the past 3 weeks.

Reminder: He has a history of type-2 diabetes mellitus to which he is non-adherent to diet or medications.

Current vital signs: His blood pressure is 140/89 mmHg, pulse is 102/min, respirations are 46/min, temperature is 104°F (40 ° C) and random blood glucose is 550 mg/dL (normal range: 70 – 110 mg/dL).

General physical examination shows a pale man with frequent coughing fits and fast labored breathing. Further examination shows swollen lymph nodes in the groin area, a perineal abscess that shows some signs of healing but is still present. Auscultation of his lungs does not show any abnormal sounds. Laboratory tests show leukocytosis, and a sputum sample is sent for culture.

Diagnosis: The physician suspects tuberculosis.

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Chronic Inflammation

- Prolonged duration **(weeks or months)** - inflammation and injury as well as attempts at repair coexist
- Causes:
  - **Persistent infection** by microorganisms that are difficult to eradicate
    - mycobacteria and certain viruses, fungi, and parasites
    - delayed-type hypersensitivity
    - granulomatous reaction
  - **Hypersensitivity diseases**
    - Excessive and inappropriate activation of the immune system
      - autoimmune diseases - rheumatoid arthritis and multiple sclerosis
      - unregulated immune responses against microbes – e.g. inflammatory bowel disease
    - The reactions are triggered against antigens that are normally harmless and serve no useful purpose
      - may show morphologic patterns of mixed acute and chronic inflammation
      - fibrosis may dominate the late stages
  - **Prolonged exposure to toxins** – endogenous or exogenous
    - silicosis or atherosclerosis

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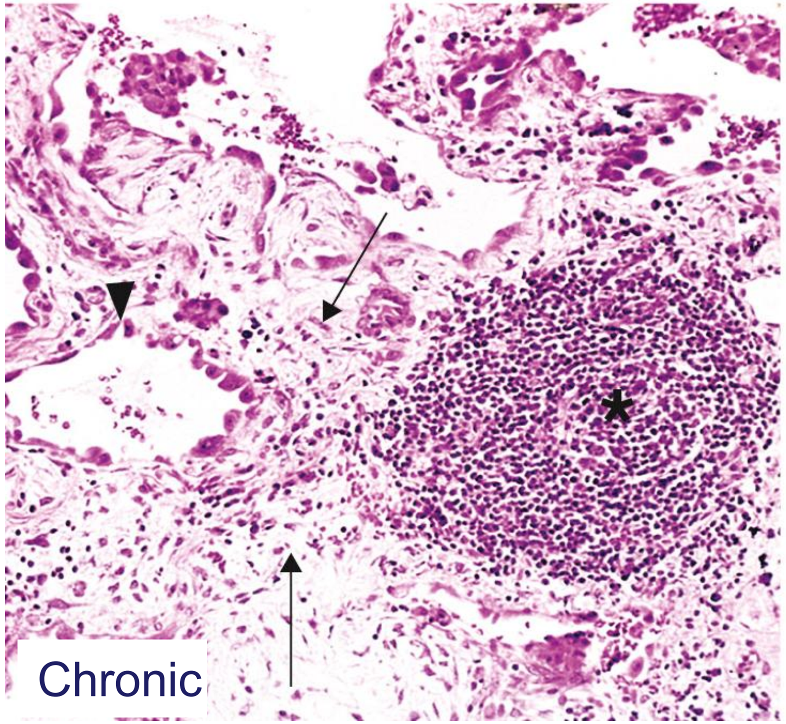
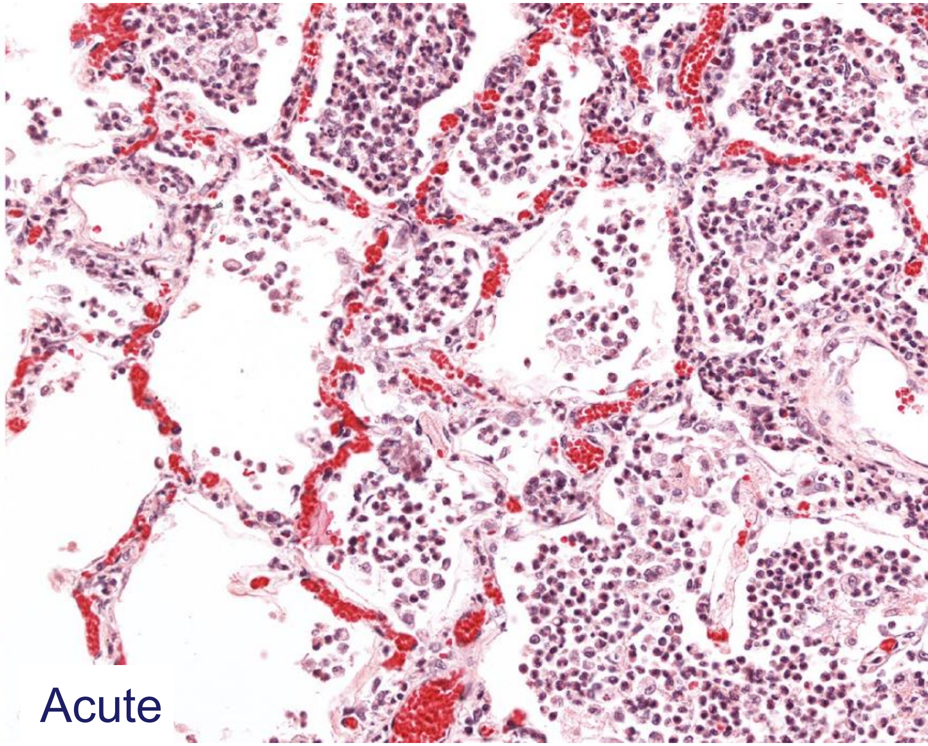
Chronic Inflammation – morphologic features

- Infiltration with **mononuclear cells** - macrophages, lymphocytes, plasma cells
- Tissue **destruction** - by the persistent offending agent or inflammatory cells
- Attempts at healing by connective tissue replacement of damaged tissue, accomplished by angiogenesis (proliferation of small blood vessels) and fibrosis.





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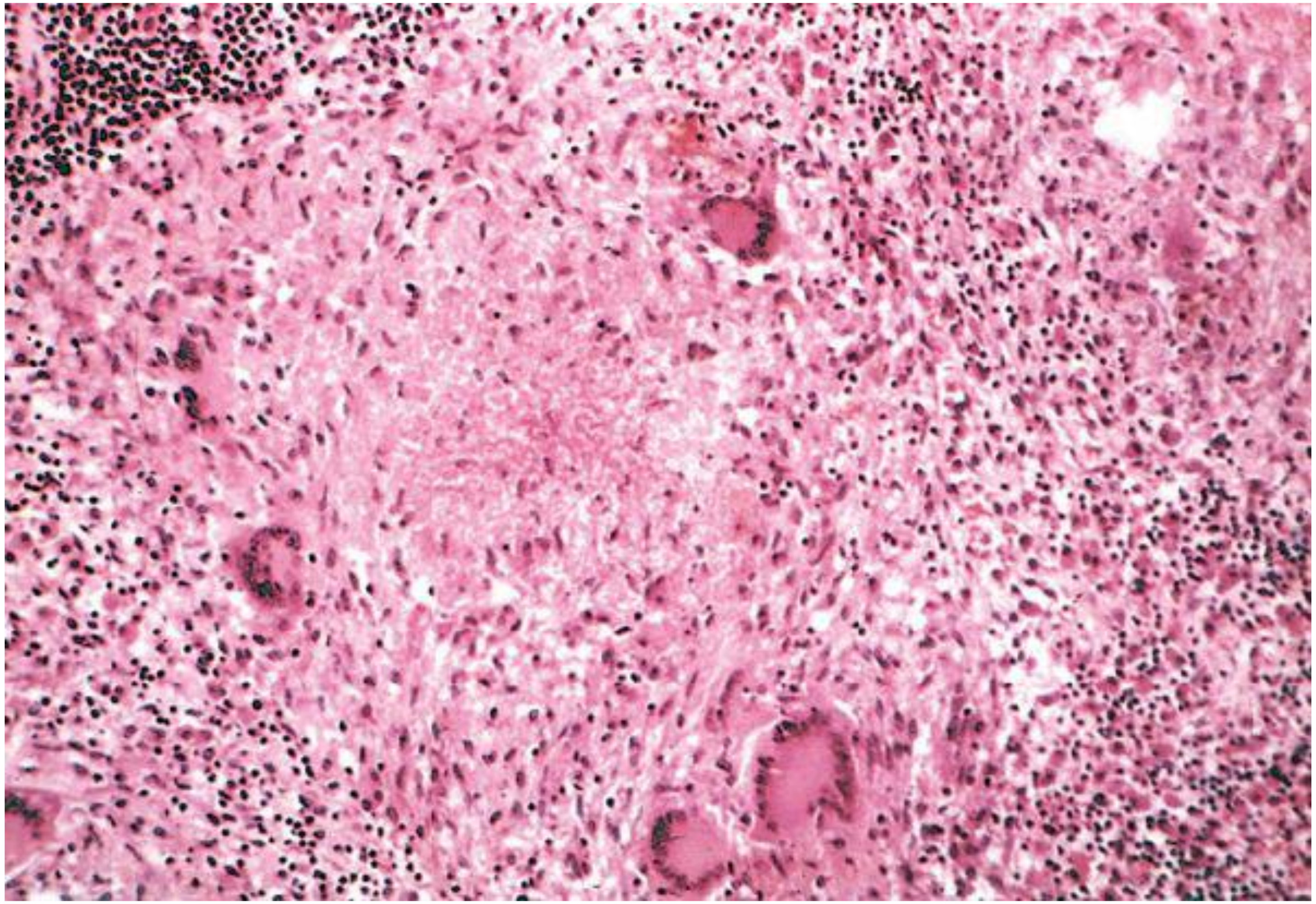
Tuberculosis Granuloma

- Cause
  - Mycobacterium tuberculosis
- Morphology
  - Caseating granuloma
  - Central necrosis
  - Epithelioid macrophages
  - Langhans' giant cells
  - Lymphocytes



Robbins 8th ed.

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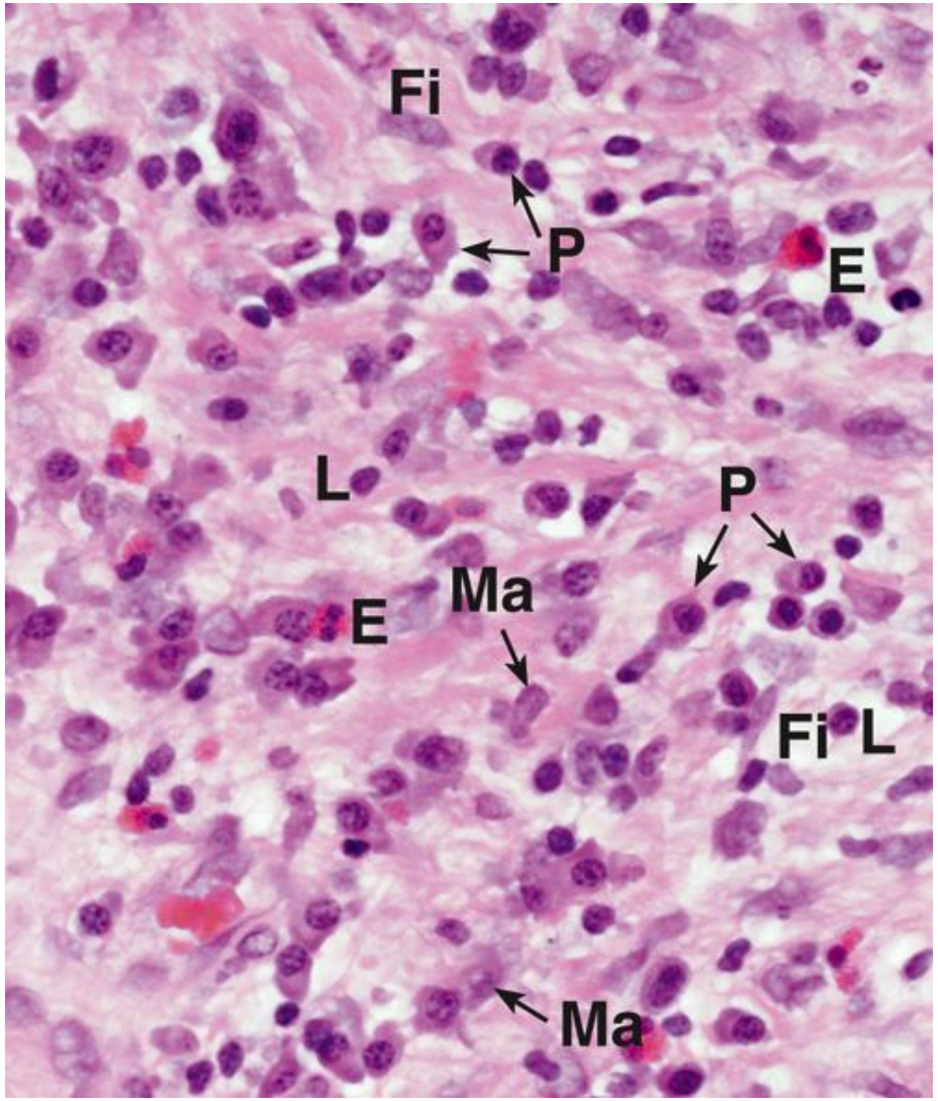
Chronic Inflammation – cells

- **Macrophages** – secrete cytokines (TNF, IL-1, chemokines, and others), eicosanoids and growth factors,
  - destroy foreign invaders and tissues, and activate other cells (T lymphocytes)
  - initiate the process of tissue repair and are involved in scar formation and fibrosis
- **Lymphocytes** - Microbes and other environmental antigens activate T and B lymphocytes, which amplify and propagate chronic inflammation
- **Plasma cells**
- **Eosinophils, mast cells & fibroblasts** (may see a few neutrophils)



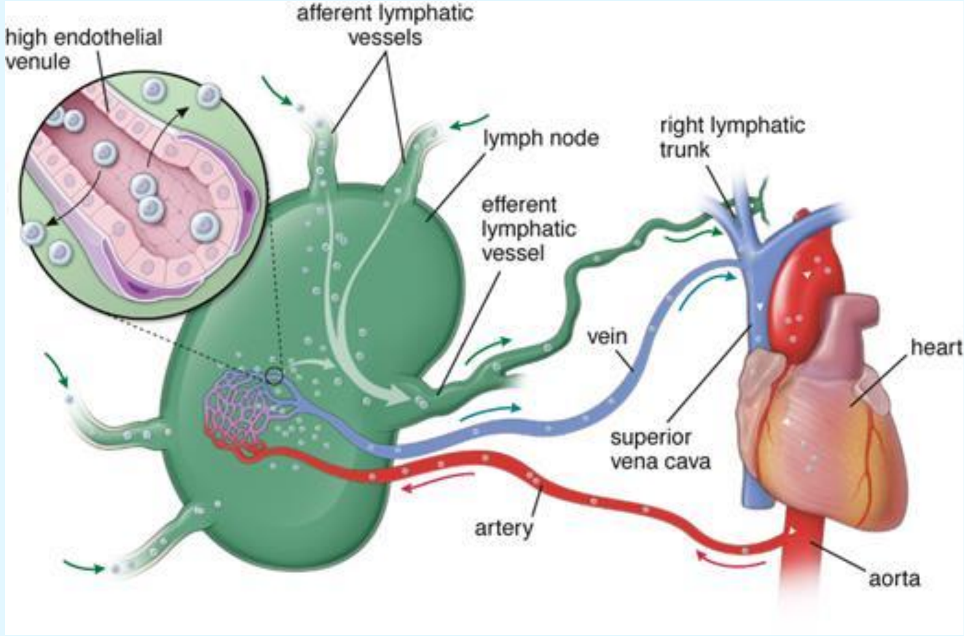
Lymphocytes and macrophages interact in a bidirectional way, and these interactions play an important role in propagating chronic inflammation

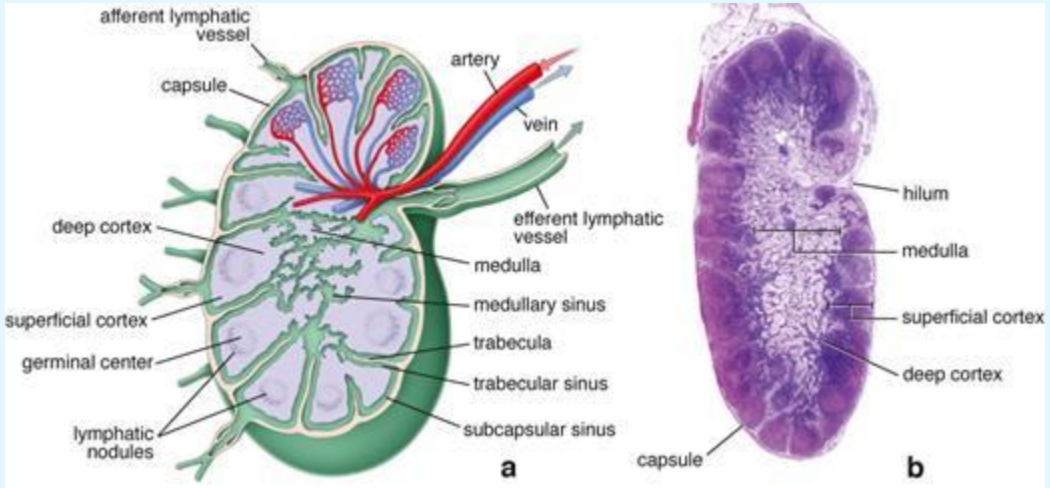
Additional things to consider

- What purpose does the anti-inflammatory medication serve in treatment of this case?
- Why were his lymph nodes swollen?
- What other tissues and organs play a role in the immune response?

Secondary lymphoid organs - Lymph nodes

- Encapsulated kidney or bean shaped structures
- Found everywhere in the body but clusters form in specific areas
  - Named nodes e.g. **axillary, inguinal**
- Lymphocytes enter and leave freely
  - **High endothelial venules** (from blood)
  - **Afferent lymphatic vessels** (from lymph)
  - In the node they occupy specific regions
  - Once activated they leave via efferent lymphatic vessels





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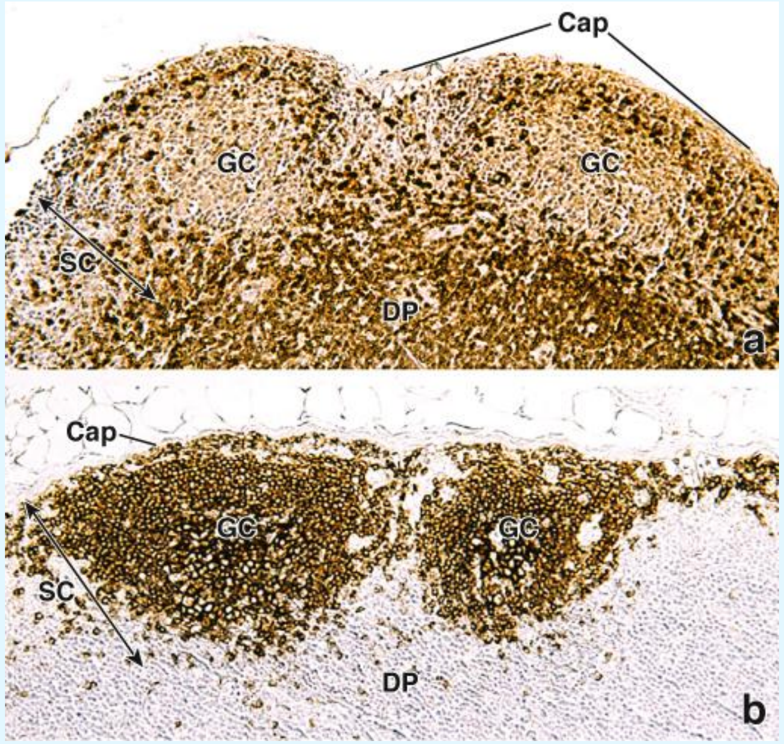
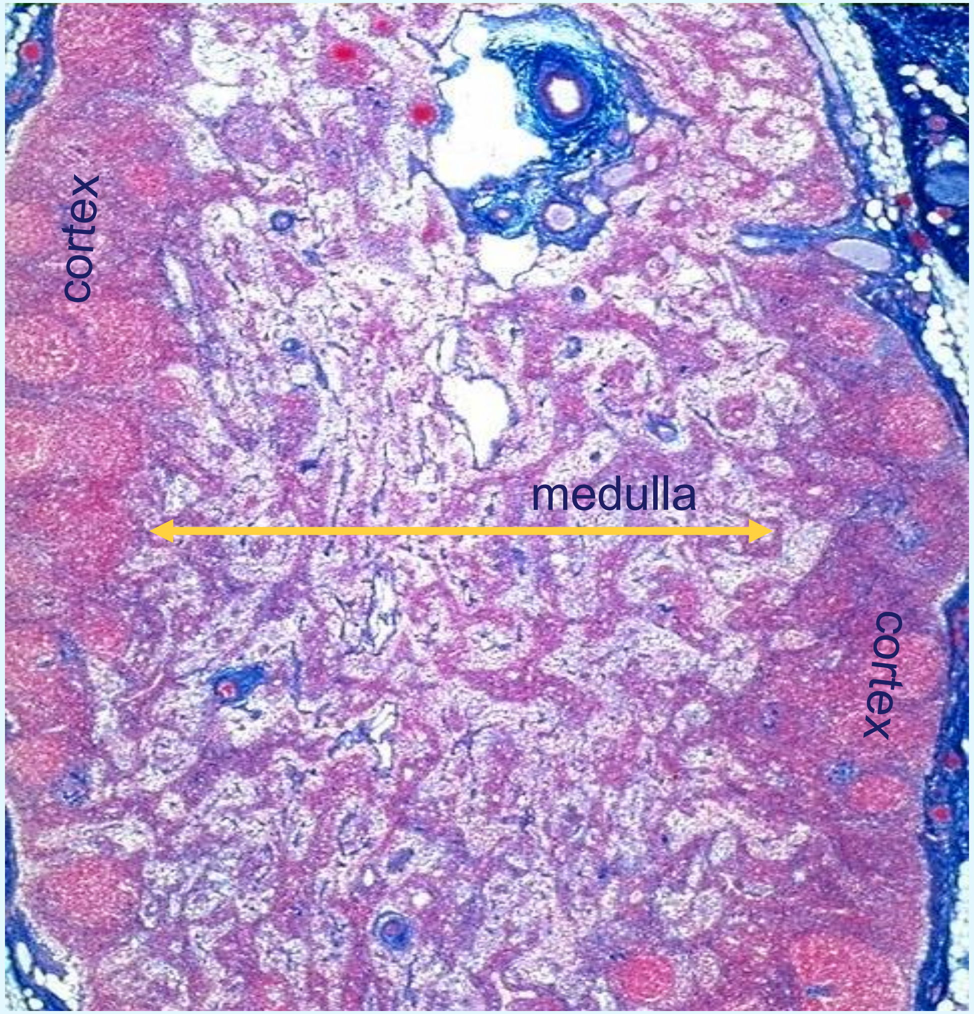
Question 19

Lymph nodes – internal architecture – superficial cortex

- Characterized by the presence of lymphatic nodules
- Primary and secondary nodules
  - **Germinal centers**
- Mainly consisting of **B lymphocytes**
- Presence of few T lymphocytes, macrophages, reticular cells and APCs (antigen presenting cells)



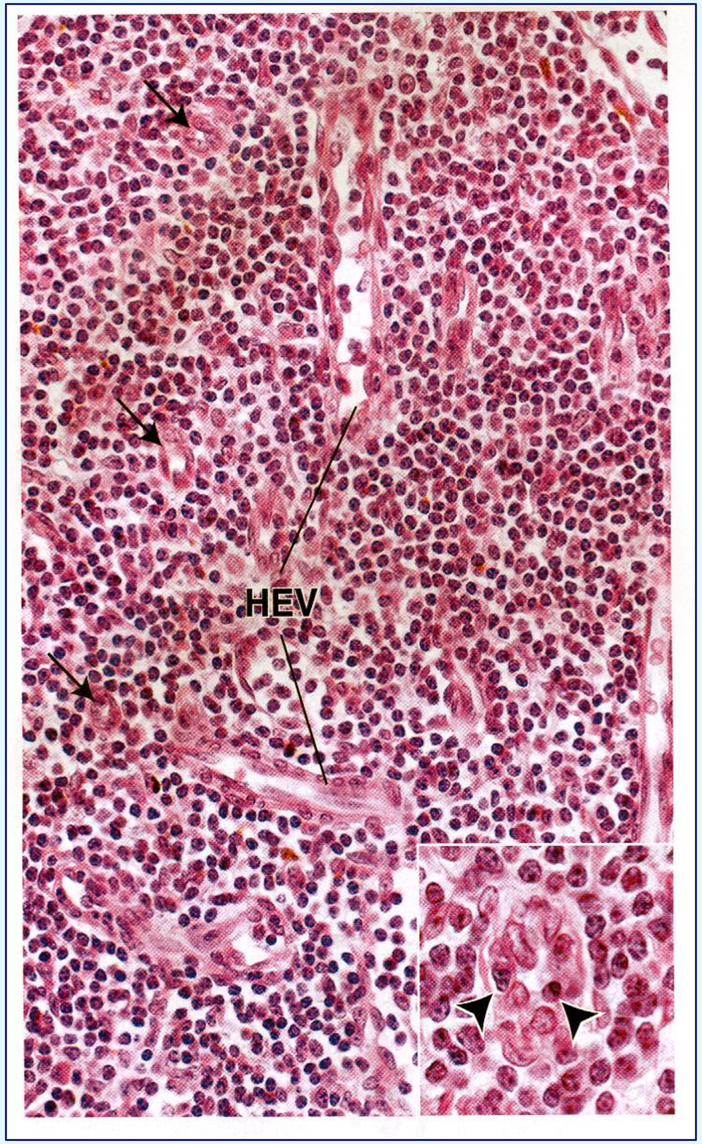


Longitudinal section through a lymph node- trichrome stain

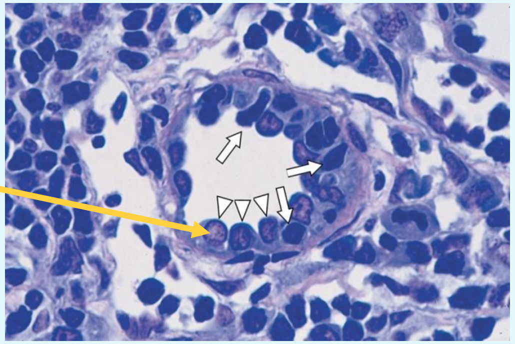
Question 19

Lymph nodes – internal architecture – deep cortex

- Primarily **T lymphocytes**
- Characterized by the presence of **high endothelial venules (HEV)**
  - allow for the transition of lymphocytes from the blood stream to lymph tissue within lymph node
- Excess lymph fluid circulating within a lymph node (average 35%) is reabsorbed into these vessels



HEV



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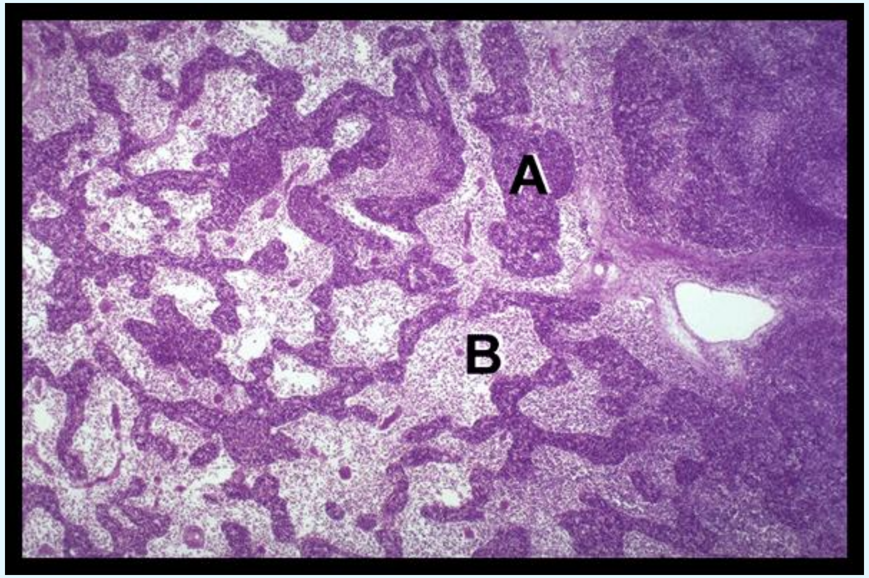
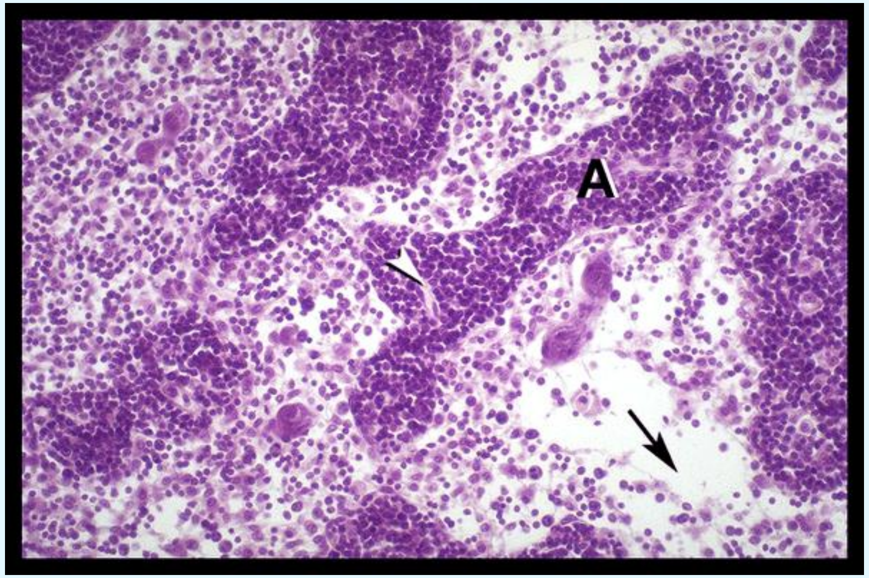
Question 19

Lymph nodes – internal architecture – medulla

- Cords (A):
  - branched cordlike extension of dense lymphoid tissue
  - contain primarily B lymphocytes, plasma cells, reticular cells and macrophages

- Sinuses (B):
  - dilated spaces separating medullary cords
  - contain lymph, few wandering macrophages (create meshwork that slows flow)
  - granulocytes maybe present when lymph node is draining an infected region





Medulla of a lymph node- high magnification- H & E stain

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Tissues and organs of the lymphatic system

- Aggregations of lymphoid accumulations = **tissues**
  - Diffuse Lymphoid Tissue
  - Nodular Lymphoid Tissue
    - Solitary Nodules
    - Aggregate Nodules
- Lymphocytes develop and mature in **primary lymphoid organs**
- **Secondary lymphoid organs** serve as collection sites where cells of the immune system can mingle